

The Kodamai Container Program: A Complete Technical Record

Part III — Differential Structure

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Abstract

Part III records the differential structure of the container program. The AAGM one-hole container derivative ∂ is the differential combinator of a Cartesian differential category, equivalently a tangent structure realised by dual-number substitution $T(p) = p(y + \varepsilon v) \bmod \varepsilon^2$. The uniqueness is peer-reviewed; the Cartan/negation boundary is the honest limit. Every result carries its trust grade verbatim and its registry node.

Grade legend. [peer-reviewed] > [proved] > [lean-verified]/[computed]. Grades are copied verbatim and never inflated.

Thesis of Part III. The AAGM one-hole container derivative ∂ is not an ad-hoc operation: it is the differential combinator of a Cartesian differential category, equivalently a tangent structure realised by dual-number substitution $T(p) = p(y + \varepsilon v) \bmod \varepsilon^2$. The uniqueness is peer-reviewed; the Cartan/negation boundary is the honest limit.

1 The container derivative is a tangent structure

Precis. SPoly (finite-support polynomial functors, substitution) is a Cartesian differential category with combinator $D = \partial$; hence a Cartesian tangent category.

Theorem 1.1 (∂ is a CDC differential combinator, [proved]). *∂ is the differential combinator of a Cartesian differential category of polynomial functors. Node `tangent-container-derivative`.*

Theorem 1.2 (Dual-number tangent, [proved]). *$T(p) = p(y + \varepsilon v) \bmod \varepsilon^2$ is a strict $\triangleleft$ -monoidal base change $\text{DCont}(k) \rightarrow \text{DCont}(D)$, $D = k[\varepsilon]/\varepsilon^2$, with the laxator equal to the chain rule on the nose. Node `tangent-restricts-to-dcont`.*

2 Uniqueness of the container tangent structure

Precis. On finite-support polynomial functors, ∂ is the unique nontrivial representable tangent structure. This is the strongest-graded differential result.

Theorem 2.1 (Uniqueness, [peer-reviewed]). *On SPoly, ∂ is the unique nontrivial representable tangent structure, in the **corrected statement** per the Rick referee (2026-10-05). Node `tangent-uniqueness-spoly`.*

Flag. State the corrected form, not the earlier unconditional slogan.

3 The Cartan / negation boundary

Precis. The full Cartan calculus requires fiberwise negation, which SPoly lacks; ring-completion $\mathbb{N} \rightarrow \mathbb{Z}$ switches it on. This is the precise edge of the differential story.

Theorem 3.1 (Negation obstruction, [proved]). *SPoly/ ∂ has no fiberwise negation $\implies$ no scalar ring object $\implies$ the Cartan calculus fails over $\mathbb{N}$. Node `cartan-scalar-object-spoly`.*

Theorem 3.2 (Ring-completion switches Cartan on, [proved]). *Fiberwise group completion $\mathbb{N} \rightarrow \mathbb{Z}$ switches the full Cartan calculus ON (SPoly $_{\mathbb{Z}}$) — the far side of the “negation knife.” Node `virtual-container-cartan`.*

Theorem 3.3 (Lie-parallelizability port, [proved]). *A poly comonoid is tangent-parallelizable $\iff$ an out-degree condition; the Lanfranchi port is ill-posed / composition-blind. Node `lie-parallelizable-poly-comonoid`.*